

Paul He

☎ 416 434 2761

✉ paul.he@mail.utoronto.ca

🌐 [linkedin](#)

🔄 GitHub

🌐 Academic Website

Education

University of Toronto

Honours Bachelor of Science in Computer Science

September 2021 - May 2025

Toronto, Canada

- Enrolled in Arts & Science Co-Op program
- Dean's List
- **Relevant Coursework:** Introduction to Machine Learning

Swiss Federal Institute of Technology (ETH Zürich)

Exchange Year

September 2023 - August 2024

Zürich, Switzerland

- EESTEC Generative AI hackathon winner
- **Relevant Coursework:** Visual Computing, Natural Language Processing (Graduate), Probabilistic Artificial Intelligence (Graduate), Machine Perception (Graduate), Large Language Models (Graduate), Advanced Formal Language Theory (Graduate)

Experience

Student Researcher

Rycolab, Institute of Machine Learning, ETH Zürich

June 2024 - Present

Zürich, Switzerland

- Engaged in cutting-edge research under the supervision of Tianyu Liu, Prof. Dr. Ryan Cotterell.
- Contributed in the design of **novel attention mechanism** incorporating concepts from **information theory**.
- Conducted experiments by implementing the proposed mechanism with **PyTorch** and training on **GPU clusters**.
- Analyzed experimental results and created synthetic data to generalize problem scenarios.
- Prepared documentation to communicate research progress.

Projects

Word Segmentation and Part-of-Speech Tagging with Transformer

October 2023 - January 2024

- Designed a **new decoder embedding mechanism** allowing the decoder to gain access to finer-grained features, enabling it to better capture subtle linguistic nuances and dependencies between characters and tags.
- Implemented a **transformer** model using **PyTorch**, beam-search for generation.
- Achieved leading-edge results, demonstrating the model's efficiency with a validation accuracy of **96.29%** for CWS, **93.37%** for POS tagging, and **91.99%** for joint CWS and POS tagging after just 10 epochs.

Mental Health Support Bot | Python, beautifulsoup4, FastAPI

October 2023

- Led the development of simple mental health chatbot web-app tailored for ETH Computer Science students, assisting team members in debugging.
- Performed **web scraping** to obtain data to perform **few-shot prompting**
- Implemented a real-time chat feature FastAPI to host our chat platform and model on a server, allowing multiple chat's with the bot at once, and also recalls previous conversations

Clinic Management System | Java, JUnit, Git

June 2022 - August 2022

- Created a Clinic Management System adhering to **Clean Architecture** and **SOLID principles**, while incorporating **Design Patterns** for optimal code structure.
- Collaborated with a **team of 8**, leveraging **Git & Github** for efficient version control, pull requests, issue tracking, and code reviews.
- Ensured code reliability through comprehensive test coverage by developing robust test suites using the JUnit framework.

Technical Skills

Programming Languages: Python, Java, C, R, SQL

Frameworks/Libraries: PyTorch, scikit-learn, NumPy, FastAPI, Flask

Tools/Platforms: Git, Docker, Jupyter, GitHub

Databases: PostgreSQL, psycopg2